Logical Operators Quick Reference				
¬	∧	∨	→	↔
∀	∃	∈	∉	≡

1A. $(2^4)(3^3)(5^3)(13^2)(23^2)$

We know:
46^2 = 2^2 x 23^2
26^2 = 2^2 x 13^2
15^3 = 3^3 x 5^5

By multiplying the common prime factors and adding their exponents, we arrive at the answer.

1B. $(2^7)(3^2)(5^1)(7^1)$

We know:
8! = 8 x 7 x 6 x 5 x 4 x 3 x 2 x 1
8! = 2^3 x 7 x 2(3) x 5 x 2^2 x 3 x 2
8! = 2^7 x 3^2 x 5 x 7

Therefore, we have arrived at our answer.

2. We know an approximation of the number of primes of a given number can be found by the following: $\pi(n) = \frac{n}{\ln(n)}$. Therefore, for the expression $8 = \sqrt[3]{n}$, we have n = 8^3 = 512. Using the formula, we have $\pi(512) = \frac{512}{\ln(512)} \approx 82$. Therefore, we have approximated the number of primes whose cube root is less than 8 to be approximately 82.

3. We know that decimal uses base 10, so there is no need to alter 321 before converting it to binary. We know the largest integer n such that 2^n < 321 is n = 8 where 2^n = 256. Therefore, we have 100000000 as the decimal representation for 256. 321 – 256 = 65. We will now repeat the process and find the largest integer n such that 2^n < 65. We find n = 6 where 2^n = 64. 65 – 64 = 1 and so we know that 2^0 = 1. Therefore, our final decimal representation is 101000001.

4A. We will first convert $(10100110)_2$ into decimal expansion. We treat the n^{th} place of the binary number as 2^n. Therefore, we have the following:

$0(2^0) + 1(2^1) + 1(2^2) + 0(2^3) + 0(2^4) + 1(2^5) + 0(2^6) + 1(2^7)$

Which is equivalent to:

$2 + 4 + 32 + 128 = 166$. Therefore, the decimal expansion – in base 10 – of $(10100110)_2$ is 166.

4B. We will convert $(10100110)_2$ into octal expansion. We will proceed by digit grouping – where we group the binary representation into groups of three starting from the right. Therefore, we have (10 100 110). We will treat each group as a binary number and convert each group into decimal representations. Therefore, we have (2 4 6). Proof? Trivial. And so, we have that the octal expansion of $(10100110)_2$ is 246.

4C. We will convert $(10100110)_2$ into hexadecimal expansion. We will proceed by digit grouping with groups of four: (1010 0110). We will convert each group into a decimal representation as used above: (10 6). Since $10 > 9$, we will find $10 \% 9 = 1$. Using the equivalency of $A = 1, B = 2, C = 3 \ldots F = 6$, we are able to convert the group decimal representation into (A 6). Therefore, we have arrived at the hexadecimal representation of A6.

5. We will convert (C9A6) into binary expansion. We will break each digit of the hexadecimal representation into an individual digit: (C 9 A 6). We know, in hexadecimal, each digit is represented by a group of four binary digits. Therefore, we find the following: (1100 1001 1010 0110) by using the process as described in 4C. Therefore, we arrive at the binary form of 1100100110100110.

6. We will convert (BA13) into octal expansion. We will break each digit of the hexadecimal representation into an individual digit: (B A 1 3). We know, in hexadecimal, each digit is represented by a group of four binary digits. Therefore, we find the following: (1011 1010 0001 0011) by using the process as described in 4C. We will now make groups of three, as described in 4B to yield the following: (001 011 101 000 010 011). Now, we will convert each group of three into the decimal form of the binary group resulting in (1 3 5 0 2 3). Therefore, we arrive at our answer of 135023.

7A. (43^2 mod 36) mod 8

We will first simplify the equation using modular properties:
$(43^2 \text{ mod } 36) \text{ mod } 8 = (43 \text{ mod } 36)^2 \text{ mod } 36 \text{ mod } 8$

Then we will solve the parentheses to obtain:
$7^2 \text{ mod } 36 \text{ mod } 8$

Solving, we get:
13 mod 8

Resulting in the final answer of:
5

7B. (9^3 mod 11)^2 mod 18

We will first simplify the equations using modular properties:
$[(9 \text{ mod } 11)^3 \text{ mod } 11]^2 \text{ mod } 18$

We will then solve the inner parenthesis to obtain:
$[8 \text{ mod } 11]^2 \text{ mod } 18$

Solving, we simplify down to:
9 mod 18

Resulting in the final answer of:
9

7C. (24^2 mod 6) mod 7003

We will first simplify the equations using modular properties:
$(24^2 \text{ mod } 6)^2 \text{ mod } 6 \text{ mod } 7003$

We will then solve the inner parenthesis to obtain:
0 mod 6 mod 7003

Which simplifies to, producing our final answer:
0

7D. ((-7)^3 mod 10)^3 mod 5

We will first simplify the equations using modular properties:
$[(-7 \text{ mod } 10)^3 \text{ mod } 10]^3 \text{ mod } 5$

We will then solve the inner parenthesis to obtain:
$[27 \text{ mod } 10]^3 \text{ mod } 5$

Which simplifies to:
$7^3 \text{ mod } 5$

Which is equivalent to:
$(7 \text{ mod } 5)^3 \text{ mod } 5$

Which is simplifies to:
$2^3 \text{ mod } 5$

Which is equivalent to:
8 mod 5

Solving provides the final answer of:
3

8A. $c \equiv_{15} 11b$

Since b and 11 have the same remainder when divided by 15 – as given – it is evident that b mod 15 is equivalent to 11 mod 15. Therefore, we may conclude that it is also the case that 11b mod 15 is equivalent to c mod 15, where we are trying to find a value of c such that the expression is true. We know with regard to *mod 15* that 11 = b. And so, by substitution, we have 11(11) mod 15 is equivalent to c mod 15. Solving, we have 121 mod 15 is equivalent to c mod 15. We know c must be in the range of $0 \leq c \leq 14$, and so we have 121 mod 15 = 1 mod 15. Thus, we have found c = 1.

8B. $c \equiv_{15} a^3 + 2b^2$

Since b and 11 have the same remainder when divided by 15 – as given – it is evident that b mod 15 is equivalent to 11 mod 15. Since a and 2 have the same remainder when divided by 15 – as given – it is evident that a mod 15 is equivalent to 2 mod 15. Therefore, we may conclude that is also the case that $a^3 + 2b^2$ mod 15 is equivalent to c mod 15, where we are trying to find a value c such that the expression is true. We know with regard to *mod 15* that 11 = b and 2 = a. And so, by substitution, we have $2^3 + 2(11)^2$ mod 15 is equivalent to c mod 15. Solving, we have 8 + 242 mod 15 is equivalent to c mod 15. By simplification, we have 250 mod 15 is equivalent to c mod 15. We know c must be in the range of $0 \leq c \leq 14$, and so we have 250 mod 15 = 10 mod 15. Thus, we have found c = 10.

8C. $c \equiv_{15} (8a)^{2000}$

Since a and 2 have the same remainder when divided by 15 – as given – it is evident that a mod 15 is equivalent to 2 mod 15. Therefore, we may conclude that is also the case that $(8a)^{2000}$ mod 15 is equivalent to c mod 15, where we are trying to find a value c such that the expression is true. We know with regard to *mod 15* that 2 = a. And so, by substitution, we have $(16)^{2000}$ mod 15 is equivalent to c mod 15. The first expression is equivalent to $(16 \text{ mod } 15)^{2000}$ mod 15 which simplifies to 1^{2000} mod 15. By simplification, 1 mod 15. Using the second expression, we have 1 mod 15 is equivalent to c mod 15. Therefore, we have found that c = 1.

9. If a divides b and b divides a, then there exists integers x and y such that b = ay and a = bx. By substitution, we have a = (ay)x = axy. By division of both sides by a, we have 1 = xy. Therefore, since x and y are integers, we have x = y = 1 or x = y = -1. Therefore, we have either b = a(1) or b = a (-1). Therefore, we have either b = a or b = -a, thereby showing that if a|b and b|a then b = a or b = -a.

10. We will list every number from 2 to 114 as seen below:
2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,104,105,106,107,108,109,110,111,112,113,114

We will then remove all multiples of 2 as seen below:
2,**3**,~~4~~,**5**,~~6~~,**7**,~~8~~,**9**,~~10~~,**11**,~~12~~,**13**,~~14~~,**15**,~~16~~,**17**,~~18~~,**19**,~~20~~,**21**,~~22~~,**23**,~~24~~,**25**,~~26~~,**27**,~~28~~,**29**,~~30~~,**31**,~~32~~,**33**,~~34~~,**35**,~~36~~,**37**,~~38~~,**39**,~~40~~,**41**,~~42~~,**43**,~~44~~,**45**,~~46~~,**47**,~~48~~,**49**,~~50~~,**51**,~~52~~,**53**,~~54~~,**55**,~~56~~,**57**,~~58~~,**59**,~~60~~,**61**,~~62~~,**63**,~~64~~,**65**,~~66~~,**67**,~~68~~,**69**,~~70~~,**71**,~~72~~,**73**,~~74~~,**75**,~~76~~,**77**,~~78~~,**79**,~~80~~,**81**,~~82~~,**83**,~~84~~,**85**,~~86~~,**87**,~~88~~,**89**,~~90~~,**91**,~~92~~,**93**,~~94~~,**95**,~~96~~,**97**,~~98~~,**99**,~~100~~,**101**,~~102~~,**103**,~~104~~,**105**,~~106~~,**107**,~~108~~,**109**,~~110~~,**111**,~~112~~,**113**,~~114~~

The red numbers above are the remaining numbers. We will then proceed to the next prime number, which is 3, and remove all multiples of 3 to produce the following:
2,**3**,~~4~~,**5**,~~6~~,**7**,~~8~~,~~9~~,~~10~~,**11**,~~12~~,**13**,~~14~~,~~15~~,~~16~~,**17**,~~18~~,**19**,~~20~~,~~21~~,~~22~~,**23**,~~24~~,**25**,~~26~~,~~27~~,~~28~~,**29**,~~30~~,**31**,~~32~~,~~33~~,~~34~~,**35**,~~36~~,**37**,~~38~~,~~39~~,~~40~~,**41**,~~42~~,**43**,~~44~~,~~45~~,~~46~~,**47**,~~48~~,**49**,~~50~~,~~51~~,~~52~~,**53**,~~54~~,**55**,~~56~~,~~57~~,~~58~~,**59**,~~60~~,**61**,~~62~~,~~63~~,~~64~~,**65**,~~66~~,**67**,~~68~~,~~69~~,~~70~~,**71**,~~72~~,**73**,~~74~~,~~75~~,~~76~~,**77**,~~78~~,**79**,~~80~~,~~81~~,~~82~~,**83**,~~84~~,**85**,~~86~~,~~87~~,~~88~~,**89**,~~90~~,**91**,~~92~~,~~93~~,~~94~~,**95**,~~96~~,**97**,~~98~~,~~99~~,~~100~~,**101**,~~102~~,**103**,~~104~~,~~105~~,~~106~~,**107**,~~108~~,**109**,~~110~~,~~111~~,~~112~~,**113**,~~114~~

We will then remove all multiples of 5 since 5 is the next prime number:
2,**3**,~~4~~,**5**,~~6~~,**7**,~~8~~,~~9~~,~~10~~,**11**,~~12~~,**13**,~~14~~,~~15~~,~~16~~,**17**,~~18~~,**19**,~~20~~,~~21~~,~~22~~,**23**,~~24~~,~~25~~,~~26~~,~~27~~,~~28~~,**29**,~~30~~,**31**,~~32~~,~~33~~,~~34~~,~~35~~,~~36~~,**37**,~~38~~,~~39~~,~~40~~,**41**,~~42~~,**43**,~~44~~,~~45~~,~~46~~,**47**,~~48~~,**49**,~~50~~,~~51~~,~~52~~,**53**,~~54~~,~~55~~,~~56~~,~~57~~,~~58~~,**59**,~~60~~,**61**,~~62~~,~~63~~,~~64~~,~~65~~,~~66~~,**67**,~~68~~,~~69~~,~~70~~,**71**,~~72~~,**73**,~~74~~,~~75~~,~~76~~,**77**,~~78~~,**79**,~~80~~,~~81~~,~~82~~,**83**,~~84~~,~~85~~,~~86~~,~~87~~,~~88~~,**89**,~~90~~,**91**,~~92~~,~~93~~,~~94~~,~~95~~,~~96~~,**97**,~~98~~,~~99~~,~~100~~,**101**,~~102~~,**103**,~~104~~,~~105~~,~~106~~,**107**,~~108~~,**109**,~~110~~,~~111~~,~~112~~,**113**,~~114~~

Next, we will remove all multiples of 7 resulting in the following:
2,**3**,~~4~~,**5**,~~6~~,**7**,~~8~~,~~9~~,~~10~~,**11**,~~12~~,**13**,~~14~~,~~15~~,~~16~~,**17**,~~18~~,**19**,~~20~~,~~21~~,~~22~~,**23**,~~24~~,~~25~~,~~26~~,~~27~~,~~28~~,**29**,~~30~~,**31**,~~32~~,~~33~~,~~34~~,~~35~~,~~36~~,**37**,~~38~~,~~39~~,~~40~~,**41**,~~42~~,**43**,~~44~~,~~45~~,~~46~~,**47**,~~48~~,~~49~~,~~50~~,~~51~~,~~52~~,**53**,~~54~~,~~55~~,~~56~~,~~57~~,~~58~~,**59**,~~60~~,**61**,~~62~~,~~63~~,~~64~~,~~65~~,~~66~~,**67**,~~68~~,~~69~~,~~70~~,**71**,~~72~~,**73**,~~74~~,~~75~~,~~76~~,~~77~~,~~78~~,**79**,~~80~~,~~81~~,~~82~~,**83**,~~84~~,~~85~~,~~86~~,~~87~~,~~88~~,**89**,~~90~~,~~91~~,~~92~~,~~93~~,~~94~~,~~95~~,~~96~~,**97**,~~98~~,~~99~~,~~100~~,**101**,~~102~~,**103**,~~104~~,~~105~~,~~106~~,**107**,~~108~~,**109**,~~110~~,~~111~~,~~112~~,**113**,~~114~~

Since the next prime number is 11 and 11^2 = 121 > 114, we will not remove all multiples of 11 since there are none left. Then the cycle stops and we have the prime numbers less than 115.

From the list, we have the following prime numbers:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113

11A. We will find the GCD of -294 and 274. First, we will determine the prime factorization of each number.

$-294 = (-1)(2^1)(3^1)(7^2)$
Above, we will assume it is possible to take the prime factorization of a negative number by simply taking the prime factorization of 294 and multiplying by -1.

$274 = (2^1)(137^1)$

Therefore, we have found the only common divisor to be 2 and thus, our GCD for -294 and 274 is 2.

11B. We will find the GCD of $2^6 3^2 11^2$ and $2^3 5^3 13$. Since the two numbers are already in prime factorization, we can simply take the common terms and their associated exponents as the GCD. Therefore, we find a GCD of $2^3 = 8$.

12A. We will find the LCM of 77 and 336 using prime factorization. We will first find the prime factorization of each number:

$77 = (7^1)(11^1)$

$336 = (2^4)(3^1)(7^1)$

We will then multiply the largest value of each term to find the LCM:

$\mathrm{LCM} = (2^4)(3^1)(7^1)(11^1) = 3696$

12B. We will find the LCM of $3^4 5^6 11^2$ and $2^4 5^3 13$. Since the two numbers are already in prime factorization, we can simply multiply the largest value of each term to find the LCM:

LCM: $(2^4)(3^4)(5^6)(11^2)(13^1) = 31853250000$

13A. We will determine if {15, 175, 39} are pairwise relatively prime by finding the prime factorization of each term:

$15 = (3^1)(5^1)$

$175 = (5^2)(7^1)$

$39 = (3^1)(13^1)$

Therefore, it can be concluded the set is pairwise relatively prime since there no element in the prime factorization that each number shares.

13B. We will determine if {63, 50, 17} are pairwise relatively prime by finding the prime factorization of each term:

$63 = (3^2)(7^1)$

$50 = (2^1)(5^2)$

$17 = (17^1)$

Therefore, it can be concluded the set is pairwise relatively prime since there no element in the prime factorization that each number shares.

14A. We will find GCD(123, 456) using the Euclid's Algorithm.

$456/123 = 3$ R 87

$123/87 = 1$ R 36

$87/36 = 2$ R 15

$36/15 = 2$ R 6

$15/6 = 2$ R 3

$6/3 = 2$ R 0

Therefore, since we arrived at our first pair where R = 0, we take the denominator of 3 to be the GCD. Therefore, GCD = 3.

14B. We will find GCD(423, 72) using Euclid's Algorithm.

$423/72 = 5$ R 63

$72/63 = 1$ R 9

$63/9 = 7$ R 0.

Therefore, since we arrived at our first pair where $R = 0$, we take the denominator of 9 to be the GCD. Therefore, GCD = 9.

15A. We will use Euler's Totient Function to calculate the number of integers that are relatively prime and less than 22. We will do this by finding the prime factorization of 22.

$22 = (2^1)(11^1)$

Since $\Phi(n)$ is multiplicative and we know if n is a prime number, then $\Phi(n) = n - 1$. Therefore, we have $\Phi(22) = \Phi(2)\ \Phi(11) = (2 - 1)(11 - 1) = 10$. Therefore, $\Phi(22) = 10$.

15B. We will use Euler's Totient Function to calculate the number of integers that are relatively prime and less than 23.

We know if n is a prime number, then $\Phi(n) = n - 1$. Since 23 is prime, we have $\Phi(23) = 23 - 1 = 22$. Therefore, $\Phi(23) = 22$.

15C. We will use Euler's Totient Function to calculate the number of integers that are relatively prime and less than 24. We will do this by finding the prime factorization of 24.

$24 = (2^3)(3^1)$

Since 2^3 is not prime, we run into some issues. HOWEVER, we are smart and we can work through this without crying. We may use formulas. These formulas are cool.

$$\Phi(24) = 24\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 8$$

Therefore, we have $\Phi(24) = 8$.